MobileForge: Annotation-Free Adaptation for Mobile GUI Agents with Hierarchical Feedback-Guided Policy Optimization

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Paper version: arXiv 2606.19930v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *human-authored tasks, expert demonstrations, and manually labeled rewards are costly to produce and quickly become outdated*

Apps evolve fast; **annotation-free** loops lack a **unified substrate** and use **coarse rewards**.

Motivation: two bottlenecks

1. **No substrate.** Tasks, rollouts, feedback: separate.
2. **Coarse signals.** Reward can't mark the right steps.
3. **Stale labels.** Human labels age fast.

Evaluation setup

AndroidWorld · 116 (in-domain)

MobileWorld GUI-only · 117 · Qwen3-VL-8B base

GUI-Owl-1.5-8B base

- **Online only**; no offline step metric.
- **Held out:** no MobileWorld data; critic: Gemini-2.5-Pro.

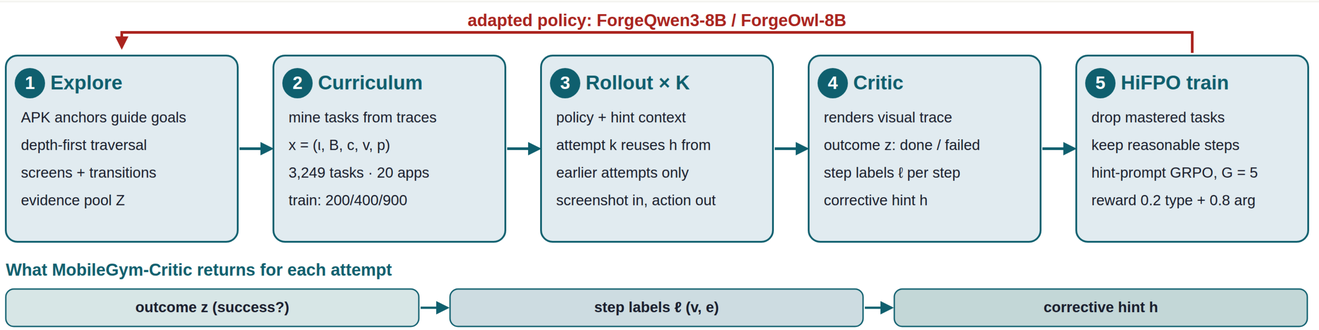
Our read (CUA Radar)

65	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out env; ablations; critic check.
- **Flags:** AP5 (data from eval benchmark); no variance; Gemini critic.
- K19 · 65 · ★★ · transfer, as recorded.

Method: MobileGym substrate + HiFPO training

- **Sees / acts:** screenshot, interaction history, prior-attempt hints in; one structured mobile action out (click, swipe, type, ...).
- **MobileGym:** explore the target app, mine trajectory-grounded tasks; the critic returns outcome z, step labels ℓ, corrective hint h.
- **HiFPO:** hints steer K attempts; drop mastered tasks, keep reasonable steps; hint-contextualized step-level GRPO.

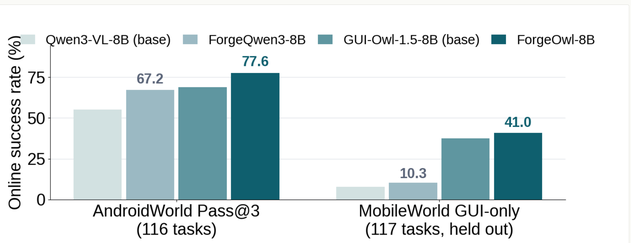


Our redraw (not the paper's figure): the annotation-free loop — explore → curriculum → hint-guided rollouts → critic → filtered step-level GRPO.

$$R_{j,g} = 0.2\,1[\hat{\alpha} = \alpha^\star] + 0.8\,1[\hat{\alpha} = \alpha^\star]\,S_{\alpha^\star}(\hat{\psi}, \psi^\star)$$

rule action reward, group-normalized inside step-level GRPO

Key results: online success



Tables 1–2, redrawn: AW Pass@3 (116) and MW GUI-only SR (117, held out); both online.

Hard-task Pass@1: 19.3% → 29.8% (GUI-Owl → ForgeOwl-8B); ForgeQwen3-8B hits 67.2% Pass@3 vs 69.0% closed-data base

Ablation: training objective

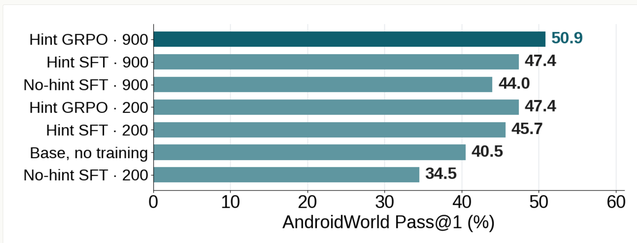


Table 8, redrawn: Qwen3-VL-8B on the same generated tasks; hint-contextualized step-level GRPO vs SFT, at 200 and 900 tasks.

SO WHAT → *No-hint SFT falls **below** the base model; hints plus step-level GRPO are strongest at both task budgets. Downstream (Table 7), removing hints drops AW Pass@1 to 49/116 (Qwen base) and 71/116 (GUI-Owl base); during rollout, hints lift success 52.0% → 77.0%.*

Headline Numbers

77.6%
AndroidWorld Pass@3
vs 69.0% base

- **41.0%** MobileWorld, held out
- **67.2%** ForgeQwen3-8B Pass@3
- **29.8%** hard-task Pass@1

The paper's own controls

CONTROL	RESULT	CONTEXT
Hints in rollout	52.0% → 77.0% success	Pass@3 49.0 → 72.5
Critic vs reference	89.75% traj · 86.53% step	Qwen cfg: 77.00/71.37
Open all-Qwen critic	AW 52/67/70 · MW 11/117	base 47/57/64 · 9/117
Task filter [0.0, 0.9)	AW 48.3% · MW 15/117	keep all-fail + mixed

Takeaway

Explore → mine tasks → hint-guided rollouts → critique → step-level GRPO: 77.6% AW Pass@3 and 41.0% held-out MobileWorld, annotation-free.

3,249 tasks · 20 apps · 8×80GB GPUs · ~80 h

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